

Notes of 'On a problem of Brown, Erdős and Sós'

Sifei Li

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1 Introduction

Let $f^{(r)}(n; s, k)$ denote the maximum number of edges in an r -uniform hypergraph on n vertices that does not contain k edges spanning at most s vertices. For integers $r > t \geq 2$ and $k \geq 2$, when $s = k(r - t) + t$, the extremal function has order n^t [1]. Several works have therefore studied the normalized quantity $n^{-t} f^{(r)}(n; k(r - t) + t, k)$ as $n \rightarrow \infty$.

The purpose of this note is to provide a complete proof of the upper bound on $n^{-t} f^{(r)}(n; k(r - t) + t, k)$ in the case where k is odd and r is large enough compared to k and t , followed by a proof sketch of Letzter and Sgueglia [3].

Theorem 1. *Let k, t, r be positive integers, k be odd, $r - 1 \geq t \geq 2$, there exists a constant $c_0(k, t)$ depending on k and t such that for all $r \geq c_0(k, t)$,*

$$\limsup_{n \rightarrow \infty} n^{-t} f^{(r)}(n; k(r - t) + t, k) \leq \frac{1}{t!} \frac{2}{2 \binom{r}{t} - 1}.$$

We may assume $k \geq 5$, as the cases $k = 1$ is trivial and $k = 3$ is settled in [2], and the argument below involves the term $k - 5$.

Notations. In an r -graph $\mathcal{F}$, we use $\mathcal{F}$ to represent the edge set $E(\mathcal{F})$, $V(\mathcal{F})$ to represent the vertex set of $\mathcal{F}$. $|\mathcal{F}|$ to represent the number of edges in $\mathcal{F}$, and $e \in \mathcal{F}$ to represent that e is an edge of $\mathcal{F}$.

For a positive integer s and k , define (s, k) -configuration to be a subgraph with k hyperedges and at most s vertices. We further shorten the notation, let k -configuration denote a $(k(r - t) + t, k)$ -configuration, a k^- -configuration denotes a $(k(r - t) + t - 1, k)$ -configuration. A hypergraph is k -free (k^- -free) if it does not contain any k -configuration (k^- -configuration).

A t -tight component is a collection of edges that can be ordered as $\{e_1, \dots, e_m\}$ such that for all i with $1 < i \leq m$, there exists an index $j < i$ such that $|V(e_j) \cap V(e_i)| \geq t$. We may use a component and a t -tight component interchangeably.

2 Preliminaries

We remark some useful results that will be used in the proof of Theorem 1.

Lemma 1 (Lemma 2.6 in [3]). *Let r, k and t be fixed positive integers. Let $\mathcal{F}$ be a k -free n -vertex r -graph. Then there exists a subhypergraph $\mathcal{F}'$ of $\mathcal{F}$ such that the following holds.*

(P1) $\mathcal{F}'$ is ℓ^- -free for every $\ell \in [2, k]$ with $\ell \mid (k - 1)$ or $\ell \mid k$;

(P2) there is no 3^- -configuration in $\mathcal{F}'$ which contains a 2-configuration;

(P3) for every positive integers a and b with $a + b = k$, every a^- -configuration and every b -configuration of $\mathcal{F}'$ are edge-disjoint;

(P4) $|\mathcal{F}'| \geq |\mathcal{F}| - O(n^{t-1})$.

Claim 1 (Claim 3.2 in [3]). Let $\mathcal{F}$ be a k -free r -graph and $e \in \mathcal{F}$. Then the number of 2-configurations of $\mathcal{F}$ containing e is at most $k - 2$.

Claim 2. Suppose that r, t, k are integers that satisfy $t \geq 2, k \geq 5$, if r is large enough such that the following inequality holds,

$$\left\{ [2r - t(2k - 3)]^t - 2[r - (k - 2)t]^t - \frac{3}{2}r^t \right\} \geq t! \left[(k - 3) + (k - 5)(k - 2) \binom{4t - 4}{t} - \frac{3}{4} \right]. \quad (1)$$

Denote $q = r - (k - 2)t, \alpha = \frac{1}{2} (2 \binom{r}{t} - 1)$. Then

$$\binom{2q}{t} - 2 \binom{q}{t} - (k - 3) - (k - 5)(k - 2) \binom{4t - 4}{t} \geq \frac{3}{2} \alpha$$

Proof. Since $\frac{n^t}{t!} \geq \binom{n}{t} \geq \frac{(n-t)^t}{t!}$, we have

$$\begin{aligned} \binom{2q}{t} - 2 \binom{q}{t} - \frac{3}{2} \binom{r}{t} &\geq \frac{1}{t!} \left\{ [2r - t(2k - 3)]^t - 2[r - (k - 2)t]^t - \frac{3}{2}r^t \right\} \\ &\geq (k - 3) + (k - 5)(k - 2) \binom{4t - 4}{t} - \frac{3}{4} \end{aligned}$$

given the assumption of r . Rearranging the equation gives the desired result. $\square$

We note that the assumption of r is feasible when r is large enough, as when $t \geq 2$, the size of the left side is $Ar^t + O(r^{t-1})$ with some positive constant A , and the right side is just a constant with respect to r . Thus, we can simply choose the constant $c_0(k, t)$ in Theorem 1 to be any number such that inequality 1 holds.

3 Upper Bound

In this section, we provide a complete proof of Theorem 1.

Proof of Theorem 1. Let F be a k -free n -vertex r -graph. By Lemma 1, $\exists \mathcal{F}' \subseteq \mathcal{F}$ such that $\mathcal{F}'$ satisfies (P1), (P2), (P3) and $|\mathcal{F}'| \geq |\mathcal{F}| - O(n^{t-1})$.

Let $\mathcal{F}_i$ be the set of edges lying in components of size i for $i = 1, 2$, and let $\mathcal{F}_3 := \mathcal{F}' \setminus (\mathcal{F}_1 \cup \mathcal{F}_2)$.

Denote $\mathcal{G}_i$ as the t -shadow of $\mathcal{F}_i$, for $i = 1, 2$. Denote $\mathcal{G}_3$ as the collection of t -sets T such that $T \notin \mathcal{G}_1 \cup \mathcal{G}_2$, and there is a unique component in $\mathcal{F}_3$ that contains a 2-configuration whose vertex set contains T .

Notice that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are disjoint. Let $\alpha = \frac{1}{2} (2 \binom{r}{t} - 1)$.

We assert that for $i = 1, 2, 3, |G_i| \geq \alpha |F_i|$. If this is true, then

$$\begin{aligned} |\mathcal{F}'| &= |\mathcal{F}_1| + |\mathcal{F}_2| + |\mathcal{F}_3| \\ &\leq \frac{1}{\alpha} (|\mathcal{G}_1| + |\mathcal{G}_2| + |\mathcal{G}_3|) \\ &\leq \frac{1}{\alpha} \binom{n}{t} \\ &= \frac{2 \binom{n}{t}}{2 \binom{r}{t} - 1} \\ &\leq \frac{2n^t}{t! (2 \binom{r}{t} - 1)} \end{aligned}$$

Therefore, $\limsup_{n \rightarrow \infty} n^{-t} |\mathcal{F}| \leq \limsup_{n \rightarrow \infty} n^{-t} (|\mathcal{F}'| + O(n^{t-1})) \leq \frac{2}{t!(2\binom{r}{t}-1)}.$

The rest of the proof is for the assertion:

For $i = 1$, by definition $|\mathcal{G}_1| = \binom{r}{t} |\mathcal{F}_1| \geq \left(\binom{r}{t} - \frac{1}{2}\right) |\mathcal{F}_1| = \alpha |\mathcal{F}_1|$, for any $r \geq t$.

For $i = 2$, for every 2-configuration (e, f) where $e, f \in \mathcal{F}_2$, we have $|V(e) \cap V(f)| \geq t$. However, since $\mathcal{F}_2$ inherits the 2^- -free property of $\mathcal{F}'$ obtained by (P1) of Lemma 1, within $2r - t - 1$ vertices one can have at most one edge, which forces $|V(e) \cap V(f)| = t$. We will reuse this property a few times in the later proof. Thus, the number of choices of t -shadow of a 2-configuration (e, f) is exactly $2\binom{r}{t} - 1$. Multiplying by the number of 2-configurations in $\mathcal{F}_2$ results in

$$|\mathcal{G}_2| = \left(2\binom{r}{t} - 1\right) \frac{|\mathcal{F}_2|}{2} = \alpha |\mathcal{F}_2|.$$

For $i = 3$, note that by definition $\mathcal{F}_3$ can be partitioned into t -tight components, each of size at least 3. Let C be a component of $\mathcal{F}_3$. It suffices to show that $|\mathcal{G}_3(C)| \geq \alpha |C|$, where $\mathcal{G}_3(C)$ denotes the restriction of $\mathcal{G}_3$ to the t -sets that are contained in the vertex sets of the edges in C . If this holds for every component, then summing over all components gives the desired result.

By the definition of a t -tight component, there are at least $|C| - 1$ distinct 2-configurations contained in C .

We first state a few observations about C that will be used later:

- C is k -free and 2^- -free, and no 3^- -configuration in C contains a 2-configuration. These properties are inherited from $\mathcal{F}$ or follow from Lemma 1.
- Each 2-configuration in C spans exactly $2r - t$ vertices, by the 2^- -free property.
- $|C| \leq k - 1$, since otherwise this would violate the k -free property.

We partition $\mathcal{G}_3(C)$ into two disjoint collections of t -sets. The first collection $\mathcal{A}$ consists of all t -sets in $\mathcal{G}_3(C)$ that are fully contained in a single edge of a 2-configuration of C , while the second collection $\mathcal{B}$ consists of all t -sets T in $\mathcal{G}_3(C)$ for which there exists a 2-configuration (e_1, e_2) in C such that $T \cap (V(e_1) \setminus V(e_2)) \neq \emptyset$ and $T \cap (V(e_2) \setminus V(e_1)) \neq \emptyset$.

We only estimate the size of $\mathcal{B}$ and show that this is enough. For each edge e in C , it shares at most t vertices with any other edge in C , since otherwise the two edges would form a 2^- -configuration. Using $|C| \leq k - 1$, it follows that there are at least $q := r - t(k - 2)$ vertices of e that are not shared with any other edge in C . We pick exactly q of them to form a subset Y_e of the vertex set.

For each 2-configuration $S = (e_1, e_2)$ in C , define $\mathcal{B}_S := \{T \subseteq Y_{e_1} \cup Y_{e_2} : |T| = t, T \cap Y_{e_1} \neq \emptyset, T \cap Y_{e_2} \neq \emptyset\}$. We have

$$|\mathcal{B}_S| = \binom{2q}{t} - 2\binom{q}{t},$$

where the term $2\binom{q}{t}$ subtracts the t -sets that are fully contained in Y_{e_1} or Y_{e_2} . It is also worth mentioning that for two distinct 2-configurations S and S' , $\mathcal{B}_S \cap \mathcal{B}_{S'} = \emptyset$.

For any $T \in \mathcal{B}_S$, if T is not contained in any edge or 2-configuration outside C , then $T \in \mathcal{G}_3(C)$. Thus we still need to remove the t -sets in $\mathcal{B}_S$ that are contained in other components. These will be considered in the following assertions:

1. The number of t -sets T in (e_1, e_2) that are contained in an edge $e' \notin C$ is at most $(k - 3)$.
2. The number of t -sets T in (e_1, e_2) that are contained in the vertex set of a 2-configuration (e', f') with $(e', f') \not\subseteq C$ is at most $(k - 2)(k - 5)\binom{4t-4}{t}$.

To prove assertion 1, suppose there is an edge $e' \notin C$ such that e' shares at least t vertices with (e_1, e_2) . Then it must share exactly t vertices with (e_1, e_2) , otherwise they would form a 3^- -configuration containing a 2-configuration, which is impossible.

For a fixed (e_1, e_2) , there are at most $(k-3)$ such edges e' , since otherwise these edges together would form a k -configuration. Therefore, the number of t -sets T in (e_1, e_2) that are contained in some edge $e' \notin C$ is at most $(k-3)\binom{t}{t} = (k-3)$.

To prove assertion 2, we first note that neither e' nor f' can intersect any edge of C in more than $t-1$ vertices, otherwise e' (or f') would lie in C , which would imply $(e', f') \subseteq C$. Thus $|V(e_1, e_2) \cap V(e', f')| \leq 4(t-1)$.

We now bound the number of such 2-configurations (e_1, e_2) . Let R be a 3-configuration in C containing (e_1, e_2) . To construct R , it suffices to find an edge $e_3 \in C \setminus \{e_1, e_2\}$ that shares at least t vertices with one of e_1 or e_2 , and set $R = (e_1, e_2, e_3)$. Such an edge e_3 always exists by the definition of a t -tight component. Indeed, ordering the edges of C as $(g_1, \dots, g_m)$ with $m \geq 3$, suppose $e_1 = g_l$ and $e_2 = g_s$ with $l < s$. If $l = 1$, then g_2 or g_3 provides a suitable choice for e_3 . If $l > 1$, then by definition of t -tightness there exists some $i < l$ such that $|V(g_i) \cap V(g_l)| \geq t$, and we may take $e_3 = g_i$.

Let $\mathcal{N}(R, C)$ be the collection of 2-configurations that are not contained in C and whose vertex sets intersect $V(R)$ in at least t vertices. Let $\mathcal{M}$ be a maximal pairwise edge-disjoint subcollection of $\mathcal{N}(R, C)$. We claim that $|\mathcal{M}| \leq \frac{k-5}{2}$.

If $|\mathcal{M}| > \frac{k-5}{2}$, choose $\frac{k-3}{2}$ members $S_1, \dots, S_{(k-3)/2}$ of $\mathcal{M}$. Then the union $R \cup S_1 \cup \dots \cup S_{(k-3)/2}$ contains $3 + 2 \cdot \frac{k-3}{2} = k$ edges, and at most $(3r-2t) + (2r-t)\frac{k-3}{2} - \frac{k-3}{2}t = k(r-t) + t$ vertices, which forms a k -configuration, contradicting the k -free assumption. Hence $|\mathcal{M}| \leq \frac{k-5}{2}$.

By maximality of $\mathcal{M}$, every 2-configuration $S \in \mathcal{N}(R, C)$ shares an edge with some member of $\mathcal{M}$. By Claim 1, the maximum number of 2-configurations containing a fixed edge is $k-2$. Thus $|\mathcal{N}(R, C)| \leq 2(k-2)|\mathcal{M}| \leq (k-2)(k-5)$.

Since each 2-configuration contributes at most $\binom{4t-4}{t}$ t -sets, we obtain the bound in assertion 2.

Therefore, each 2-configuration in C contributes at least

$$\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t}$$

distinct t -sets to $\mathcal{B}$.

Multiplying by the total number of 2-configurations in C , we have

$$\begin{aligned} |\mathcal{G}_3(C)| &\geq (|C| - 1) \left[\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t} \right] \\ &\geq (|C| - 1) \frac{3}{2}\alpha && \text{by Claim 2} \\ &= |C|\alpha + \frac{1}{2}\alpha|C| - \frac{3}{2}\alpha \\ &\geq |C|\alpha && \text{as } |C| \geq 3 \end{aligned}$$

as desired. □

4 Reference

References

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- [2] Stefan Glock, Felix Joos, Jaehoon Kim, Marcus Kühn, Lyuben Lichev, and Oleg Pikhurko. On the $(6, 4)$ -problem of brown, erdős, and sós. *Proceedings of the American Mathematical Society, Series B*, 11(17):173–186, 2024.
- [3] Shoham Letzter and Amedeo Sgueglia. On a problem of brown, erdős and sós. *Proceedings of the American Mathematical Society*, 153(07):2729–2743, 2025.